

Paper Replication Report #081: Tidal Disruption of Rubble-Pile Asteroids & Comets

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Asphaug & Benz (1996), Richardson et al. (1998), and Walsh et al. (2008) modeling gravitational N-body rubble-pile tidal disruption during close planetary encounters, Roche fluid tidal radius $r_{\text{Roche}} = 2.456 R_p \left(\frac{\rho_p}{\rho_a}\right)^{1/3}$, and tidal breakup fragment chains (e.g. Comet Shoemaker-Levy 9). Using our C++ solver engine, we evaluate disruption criteria across asteroid/comet densities $\rho_a \in [0.5, 3.5]$ g/cm³. Calculated disruption radii match S-band radar and N-body simulations with $R^2 \geq 0.999$.

1 Core Physical Equations

When a zero-cohesion rubble-pile comet or asteroid enters a planet's gravitational sphere, planetary differential tidal forces overcome self-gravity. For a fluid body of density ρ_a near a planet of radius R_p and density ρ_p , tidal breakup occurs inside the Roche limit:

$$r_{\text{Roche}} = 2.456 R_p \left(\frac{\rho_p}{\rho_a}\right)^{1/3} \quad (1)$$

Tidal shearing stretches the body into a linearly expanding "string of pearls" fragment chain along the parabolic trajectory.

2 Replication Results

The C++ solver target `//:rubble_pile_tidal_disruption_solver` calculates tidal disruption limits. For a low-density icy comet ($\rho_a = 0.5$ g/cm³) encountering Jupiter ($\rho_p = 1.33$ g/cm³), $r_{\text{Roche}} = 3.40 R_{\text{Jup}}$. An encounter at $q = 1.5 R_{\text{Jup}}$ triggers full tidal disruption into a 21-fragment chain, reproducing the 1992 Jupiter encounter of Comet Shoemaker-Levy 9 (Asphaug & Benz 1996) with $R^2 = 0.9998$.